Lab 7

Write a program to solve the following water jug problem using the approach of constraint satisfaction problem.

Example problem

Problem Definition:

- Jug A Capacity = 5 liters
- Jug B Capacity = 3 liters
- Initial State = (0, 0)
- Goal State = Jug A contains 1 liter → (1, _)

Constraints:

- Jug A can hold max 5 liters: $0 \leq a \leq 5$
- Jug B can hold max 3 liters: $0 \leq b \leq 3$
- Valid operations:
    - Fill either jug to its max
    - Empty either jug
    - Pour from one jug to another without spilling

Pseudocode

```
Function is_goal(state):
    Return True if state.A == 1

Function get_next_states(state):
    Let (a, b) = state
    max_a = 5
    max_b = 3
    Initialize empty list: successors

    # Constraint-Based Actions
    Add (5, b) // Fill Jug A
    Add (a, 3) // Fill Jug B
```

```
    Add (0, b) // Empty Jug A
    Add (a, 0) // Empty Jug B
    pour = min(a, max_b - b)
    Add (a - pour, b + pour) // Pour A → B

    pour = min(b, max_a - a)
    Add (a + pour, b - pour) // Pour B → A

    Return successors


Function water_jug_bfs(start_state):
    Initialize visited_set = empty set
    Initialize queue with (start_state, path_list)

    While queue is not empty:
        Pop (current_state, path) from queue

        If is_goal(current_state):
            Return path

        For each next_state in get_next_states(current_state):
            If next_state not in visited_set:
                Add next_state to visited_set
                Enqueue (next_state, path + [next_state])

    Return None // If no path found

Main:
    Call water_jug_bfs(start_state = (0, 0))
    If solution found:
        Print step-by-step jug contents to goal state
    Else:
        Print "No solution found"
```

Report format

Title: Write a program ……..
Date: Lab assigned date

Objective

- To implement the water jug problem solution using constraint satisfaction problem approach

Theory

- Constraint satisfaction problem
    - Water jug problem

Implementation

- Write down the problem statement
- Python Code
- Output (Screenshots of your output)

Conclusion